

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a85c569dd88172e2e.jsonl`  
- SHA-256: `92aba8fef83fef4452a57d1e59b42e0a45f98cf0131c216600eb71d18edefa75`  
- Source modified: `2026-06-06T17:32:23+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:30:48.218Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** <https://www.mdpi.com/1424-8220/24/13/4372>

****Title:**** Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera (Sensors 2024)

****Found via:**** recent/applied search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:51.273Z)

I'll fetch the source page and extract key claims relevant to the research question.

3. user (2026-06-06T17:30:59.844Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

4. assistant (2026-06-06T17:31:02.737Z)

The MDPI page returned 403. Let me try alternative access routes.

5. user (2026-06-06T17:31:05.626Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

6. user (2026-06-06T17:31:07.440Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://doi.org/10.3390/s24134372>

Redirect URL: <https://www.mdpi.com/1424-8220/24/13/4372>

Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://www.mdpi.com/1424-8220/24/13/4372"
- prompt: "Extract detailed information about this paper on badminton shot refinement via fusion of shuttlecock tracking and hit detection. Report: (1) the exact hit-detection method and inputs (audio? trajectory? pose?); (2) the shuttlecock tracking model used (TrackNet variant? WASB?); (3) how the fusion works and whether it relates to rally/point/shot temporal segmentation; (4) any accuracy/precision/recall/F1 numbers with exact figures; (5) dataset and its source/license; (6) compute/hardware requirements; (7) publication date and authors. Quote exact sentences verbatim."

7. assistant (2026-06-06T17:31:09.954Z)

MDPI is blocking WebFetch (403). Let me try fetching the page directly via curl with a browser user agent, and also search for a cached/secondary version.

8. user (2026-06-06T17:31:12.640Z)

```
EXIT:0
410 /tmp/mdpi_paper.html
<HTML><HEAD>
<TITLE>Access Denied</TITLE>
</HEAD><BODY>
<H1>Access Denied</H1>

You don't have permission to access
"http&#58;&#47;&#47;www&#46;mdpi&#46;com&#47;1424&#45;8220&#47;24&#47;13&#47;4372&#47;htm" on
this server.<P>
Reference&#32;&#35;18&#46;7176cd17&#46;1780767073&#46;15b4a1bc
<P>https&#58;&#47;&#47;errors&#46;edgesuite&#46;net&#47;18&#46;7176cd17&#46;1780767073&#46;15b4a
1bc</P>
</BODY>
</HTML>
```

9. assistant (2026-06-06T17:31:15.630Z)

MDPI is hard-blocking (Akamai edge). Let me try the PMC/PubMed mirror and Google Scholar cache, since MDPI Sensors papers are often mirrored on PubMed Central.

10. user (2026-06-06T17:31:25.977Z)

Web search results for query: "Enhancing Badminton Game Analysis Shot Refinement Fusion Shuttlecock Tracking Hit Detection Monocular Camera Sensors 2024"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/37233318/"}, {"title": "Fine reconstruction of badminton swing dynamic trajectory assisted by event camera | Scientific Reports", "url": "https://www.nature.com/articles/s41598-026-46443-8"}, {"title": "(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"}, {"title": "A Model for Analyzing Shuttlecock Landing Position Using Real-Time Object Detection Techniques | Journal of Applied Science and Emerging Technology", "url": "https://ph01.tci-thaijo.org/index.php/JASCI/article/view/265289"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "Use predictive analytics and big data technologies to enhance badminton game strategy development and performance indicators - Xin Feng, Xiang Liu, 2025", "url": "https://journals.sagepub.com/doi/10.1177/14727978251324125"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}]

I found comprehensive search results for your query! Here's what the research shows:

Study Overview

A novel method has been proposed to extract shots in badminton games from a monocular camera. This research was conducted by Hsu, Y.-H., Yu, C.-C., & Cheng, H.-Y. and published in 2024.

Methodology

The approach uses two main deep learning components:

1. **Shuttlecock Tracking**: TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock.
2. **Hit Detection**: The YOLOv7 model is used to identify whether the player is swinging.
3. **Shot Refinement**: A shot refinement algorithm is proposed to obtain the correct hitting moment, as both TrackNet and YOLOv7 may have detection misses and false detections.

Performance Results

The proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of the Badminton World Federation (BWF) match videos. This is a significant improvement compared to the use of TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score.

The research was published in the journal **Sensors** (2024, volume 24, issue 13, article 4372).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:31:28.419Z)

There's a PMC mirror (open access). Let me fetch the full text from PubMed Central, which won't be blocked.

12. user (2026-06-06T17:31:45.808Z)

Badminton Shot Refinement via Shuttlecock Tracking & Hit Detection

1. Hit Detection Method & Inputs

Method: Visual-based swing detection using YOLOv7, not audio or trajectory alone.

> "the YOLOv7 model is used to identify whether the player is swinging"

The system detects player swing actions as confidence scores:

> "a threshold is applied to the prediction outputs of the model, where a high confidence score in a specific frame implies that a player is performing a hit"

2. Shuttlecock Tracking Model

TrackNet is the backbone tracking system:

> "this study adopts the TrackNet model...specifically designed for tracking the trajectory of fast-moving balls with small volumes"

TrackNet uses CNN with VGG16 backbone and generates pixel-level probability maps across consecutive frames.

3. Shot Refinement Algorithm (Fusion & Segmentation)

The Shot Refinement Algorithm (SRA) fuses two detection streams to refine hit moments:

> "The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment"

It segments video into Hit Segments (HS) by player ID from action detection, then applies five case-based logic rules:

```
> "Case 1: Only one HD-T in HSi(k)...set HD-Tt as HMt(k)"
> "Case 4: No HD-T in HSi(k)...the frame exhibiting the highest confidence score within HSi(k)
is selected as HMt(k)"
```

This detects **hit moments within rallies**, not rally segmentation.

4. Quantitative Results (Exact Figures)

Proposed SRA vs. TrackNet baseline (IoU threshold 0.5):

Metric	TrackNet	SRA
Precision	0.588	0.843
Recall	0.936	0.882
F1 Score	0.723	0.862

Tuned HD-T + SRA (best result):

```
> "precision of the _HD-T_ (tuned) decreased to 0.524...but with the help of SRA, the precision
bounced back to 0.897"
```

Shot type classification accuracy:

- IoU 0.5: 54.1% (SRA) vs 38.8% (TrackNet)
- IoU 0.95: 72.1% (SRA) vs 71.9% (TrackNet)

5. Dataset Source & Composition

Two datasets combined:

```
> "69 matches, with 1582 rallies of the Badminton World Federation (BWF) match videos"
```

Full combined dataset:

```
> "98,675 images from 354 matches and 6855 rallies...16 players, with a gender distribution of 7
females and 9 males"
```

Sources: BWF YouTube videos + National University Competition (Taiwan Ministry of Education)

```
> "Data were obtained from https://aidea-web.tw (accessed on 29 May 2024)"
```

6. Compute Requirements & Inference Speed

Hardware:

```
> "10th generation Core i9 Intel CPU with 32 GB of RAM and an RTX 3080Ti GPU"
```

Processing time:

```
> "Processing Time (Seconds)...0.174" (SRA vs 0.032 for TrackNet alone)
```

Implementation: Python 3.8, TensorFlow 2.12

7. Publication & Authors

Published: July 5, 2024

Authors & Institution:

```
> "Yi-Hua Hsu...Chih-Chang Yu...Hsu-Yung Cheng...Department of Information and Computer
Engineering, Chung Yuan Christian University, Taoyuan City 320314, Taiwan"
```

Journal: *Sensors (Basel)*

```
> "Sensors (Basel). 2024 Jul 5;24(13):4372. doi: 10.3390/s24134372"
```

13. assistant (2026-06-06T17:31:51.996Z)

I have comprehensive extraction from the PMC open-access mirror. Let me verify a couple of the load-bearing details (the hit-detection method specifics and the F1/precision figures) by fetching the methods/results sections directly, to ensure quote accuracy.

14. user (2026-06-06T17:32:00.013Z)

Verbatim Quotes from Abstract and Conclusion

(a) Overall Performance Metrics

****Proposed Method (69 matches, 1582 rallies):****

"Our proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5%"

****TrackNet Alone:****

"This is a significant improvement compared to the use of TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score."

(b) Technical Components

****YOLOv7 for Action Detection:****

"Second, the YOLOv7 model is used to identify whether the player is swinging."

****TrackNet for Shuttlecock Tracking:****

"First, TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock."

15. assistant (2026-06-06T17:32:07.111Z)

I now have verified, verbatim claims from the open-access PMC mirror of this MDPI Sensors 2024 paper. This is primary peer-reviewed research. I have enough to extract the falsifiable claims relevant to the research question.

16. user (2026-06-06T17:32:18.524Z)

Structured output provided successfully

17. assistant (2026-06-06T17:32:23.744Z)

Extracted and verified the claims from the open-access PMC mirror (PMC11244353) since the MDPI page itself hard-blocks automated fetches (403/Akamai). Source quality is primary (peer-reviewed Sensors 2024, Chung Yuan Christian University, Taiwan). Key takeaway for the research question: this paper validates a fully-local, vision-based hit-detection family (YOLOv7 swing detection fused with TrackNet trajectory) that lifts shot-detection F1 from 0.723 to 0.905 on a single consumer GPU ? directly relevant as a complementary signal to our trajectory-only WASB windows, though it targets per-shot hit moments rather than rally start/end boundaries.